



Samurai helmet
Most battles at sea are won by selfless teamwork, but the medieval samurai warrior was able to show off his individual prowess, especially his skill as an archer, just as well on water as on land.

complement of soldiers—comparable to the Greek trireme in concept. Paddle-wheel craft, initially introduced in the 8th century, became of paramount importance under the Song dynasty. The wheels were driven by treadmills inside the hull typically operated by the leg-power of crews of 28 to 42 men. Large vessels might have 23 wheels—11 on each side and one at the stern—and measure up to 360 feet (110 m) in length. One type, known as a "sea-hawk" ship, had a low bow and a high stern, a ram at the prow and iron plates for armored protection. Used on rivers and lakes, the paddle-wheel craft were extremely maneuverable, capable of traveling forward or backward with equal ease. On-board weaponry ranged from crossbows and lances to catapults and, later, primitive cannon. Gunpowder became a common element in missile warfare in the Song period. It could be wrapped in small packages around arrowheads to make fire-arrows, or used to fuel a fire-lance—a kind of proto-flamethrower—or made into explosive grenades or bombs. Many Chinese naval battles were decided by ships being set on fire.

MING SEAPOWER

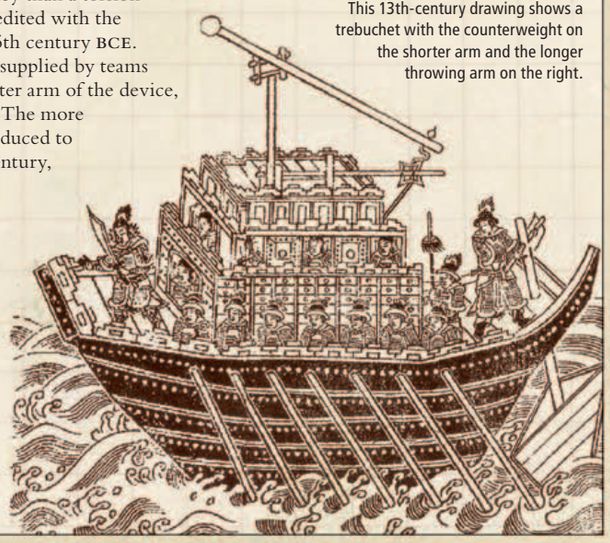
Naval conflict played a critical role in the warfare that ended in the establishment of the Ming dynasty in China in the 14th century. There were engagements between probably some of the numerically largest fleets ever assembled. Once in power, the Ming at first devoted vast resources to shipbuilding. Since they also had the magnetic compass at their disposal, there was nothing to stop them embarking upon oceanic voyages. Between 1405 and 1433 fleets led by the eunuch Admiral Zheng He cruised around southeast Asia and across the Indian Ocean, entering the Red Sea and voyaging down the east African coast as far as Mozambique. The greatest of the Chinese fleets comprised 63 large war junks—the largest more than 390 ft (120 m) long and almost 160 ft (50 m) broad—plus more than 200 support vessels and nearly 30,000 men. This constituted a thoroughly intimidating assertion of imperial suzerainty. On the few occasions when he encountered resistance, Admiral He did not hesitate to use force to impose respect—for example arresting and executing a Sumatran ruler who objected to paying tribute.

From the 1430s, however, China's Ming emperors turned instead to a policy of inward-looking development that rejected all overseas ventures. The maintenance of an ocean-going navy was completely abandoned. By the time European mariners arrived in Chinese and Japanese waters in the 16th century, they encountered no serious competitors in their bid for control of the world's ocean trade routes.

WEAPONS AND TECHNOLOGY

SONG CATAPULT SHIP

A trebuchet is a sling for hurling missiles, using the principle of the lever to give greater power and accuracy than a torsion device such as a ballista. The Chinese are credited with the invention of the first trebuchets around the 5th century BCE. The power for these traction trebuchets was supplied by teams of men pulling on ropes attached to the shorter arm of the device, the sling being at the end of the longer arm. The more powerful counterweight trebuchet was introduced to China from Muslim west Asia in the 13th century, first playing a crucial role in the siege of Xiangyang in 1273. In the Song dynasty navy trebuchets were habitually deployed as shipborne artillery, mounted on the larger "tower ships" both as a weapon for ship-to-ship combat and as a siege weapon for bombarding land fortifications from the water. When attacking ships, the trebuchets would often hurl explosive incendiary devices, an effective tactic against highly flammable vessels. Many of the Song catapult ships were driven by paddlewheels, but they could also be propelled by oars.



River ship with trebuchet
This 13th-century drawing shows a trebuchet with the counterweight on the shorter arm and the longer throwing arm on the right.